Chapter 1

From Empirical Percentiles to Hypothesis Testing: Validating the Selection of Least-Contaminated Reference Sites

1.1 Introduction

Sediment contamination poses chronic threats to benthic communities in aquatic ecosystems, and the assessment of chemical stress requires careful selection of ecologically relevant variables. The 16 chemical variables retained in this study were chosen by ecologists and toxicologists and justified from an ecological and toxicological standpoint, reflecting the judgement that each variable carries meaningful stress information for the communities under study.

Despite their distinct contamination roles, these 16 variables do not behave as equally distinct statistical entities. As random variables measured across sampling sites, many of them exhibit strong co-variation in concentration, reflecting shared contamination sources, co-deposition pathways, or common geochemical associations, and some show highly similar distributional patterns across sites. This statistical redundancy means that treating the 16 raw chemical measurements as independent lines of evidence would be misleading: correlated variables would carry overlapping information and receive collective weight, since the variance of their combined contribution exceeds the sum of their individual variances once their covariance is positive [1]. As a result, the underlying structure of contamination would be obscured rather than clarified. This raises a central methodological question: how can we extract a smaller set of latent stressors that are statistically orthogonal to one another while faithfully preserving the original contamination stress information? The value of identifying such latent stressors is: orthogonality ensures that any contamination stress signal attributed to one latent stressor is uncorrelated with all others. Then the contamination information retained within each latent stressor refers exclusively to its own domain, without carrying confounded information from the others.

With these orthogonal latent stressors, we adopt a specific contamination–community response perspective to synthesise stressor values into an integrated stress level for sites. This integrated stress level identifies sites whose communities are least affected by chemical contamination. Here it is important to distinguish two related but non-equivalent notions. A natural site refers to a community shaped under undisturbed or minimally disturbed conditions — a theoretical reference state representing biological integrity in the absence of human stressors. A least-polluted site, by contrast, is an operational concept defined relative to the observed distribution of integrated chemical stress within the study area, with sites at the low end of that distribution taken as the least-polluted.

We argue that, in this study, the least-polluted-site concept is the safer and indeed the only defensible choice of the author's baseline. As studies (Stoddard et al., 2006 [13], Davies and Jackson, 2006 [5]) emphasise in developing the Biological Condition Gradient, truly natural conditions are difficult to observe directly: in extensively altered regions, practitioners tend to accept the "best of what remains" as the apparent optimum, because the original natural state can no longer be visualised. Moreover, they note that least-disturbed reference sites differ markedly across regions in their degree of departure from historical or natural condition. Direct observation of a nat-

[1] For two recorded measures treated as random variables X and Y, the variance of their sum is $\mathrm{Var}(X+Y) = \mathrm{Var}(X)+\mathrm{Var}(Y)+2\,\mathrm{Cov}(X,Y)$. When $\mathrm{Cov}(X,Y) > 0$, this exceeds $\mathrm{Var}(X)+\mathrm{Var}(Y)$, so the shared covariance is absorbed into the combined variance and cannot be attributed to X or Y individually.

ural community is therefore not attainable from the available data; the least-polluted sites, identified from the low tail of the integrated stress distribution, serve as the available approximation to natural community organisation.

The ecological significance of these least-contaminated sites follows directly. Within them, community structure is expected to be governed primarily by habitat and environmental factors rather than by chemical stress, so they retain the information of naturally organised communities. These natural community information are comparable benchmark — both for understanding natural community composition and for studying the community changes induced by contamination stress.

The central contribution of this chapter is methodological. Rather than delineating the least-contaminated group by a fixed empirical percentile—a convention widely adopted–we defend the delineation threshold as a question to be tested. Using two complementary resampling-based hypothesis tests, we ask whether a candidate reference group of a given size genuinely carries weaker contamination-constrained community variation than would be expected by chance, and we use this evidence to support a selected least-contaminated set rather than empirically assuming it. By generalizing this approach to different study cases, it provides informative guidance that least-contamined sites need to be set according to the observed contamination stress and not by a fixed empirical percentile.

1.2 Methods

1.2.1 Principle Component Analysis/R-model Ordination on Chemical Matrix

In linear algebra, the eigen-decomposition of a matrix identifies a set of characteristic directions along which a linear transformation acts purely by scaling, and within a measure space it resolves the structure of that space into mutually orthogonal axes whose orthogonality can be formally proven from the symmetry of the underlying matrix (Strang, 2006 [14]). When this decomposition is applied to a data matrix, it acquires a statistical interpretation: Principal Component Analysis is an eigen-decomposition performed after a descriptive space — conventionally the column space — has been chosen for the data matrix, and in numerical ecology it is classified among the R-mode ordination methods that operate under a Euclidean distance measure of dissimilarity (Borcard, Gillet, and Legendre, 2018 [4]). PCA extracts the eigenvectors from the correlation matrix of the column space, and the resulting orthogonal eigenvectors are scaled so as to preserve, as far as possible, the structure and magnitude of the data within the original descriptive space. The values of the original variables are then projected onto these orthogonal vectors and come to lie in a new descriptive space spanned by the ordination axes, the principal components. It must be emphasised that orthogonality holds only among the newly constructed principal components; the covariance among the original variables is still partly retained within these components. This is reflected in the fact that the variance of each principal component preserves not only the original quantity of information (variance) but also the covariance information among the original variables.

In this study, the structure and magnitude of the variance carried by the principal components themselves are not our central concern. What concerns us is rather the assurance that the principal components preserve the original information content together with their mutual orthogonality, that is, their statistical independence. In this way, when we discuss the contamination stress information of one or several stressors at a given site, we can ensure that it is uncorrelated and completely retained within the domain to which the latent stressor(s) refer, allowing the contamination variation arising from different latent stressors to be correctly distinguished.

A brief proof of the mathematical properties underlying PCA is provided in the Appendix 1.5.1). Performing PCA on the chemical matrix, rotating the loadings, flipping to ensure most positive loadings across chemicals, and selecting the leading principal components, yields a set of latent stressors that quantify the contamination stress information. The full procedure, from the input chemical matrix to the output latent stressors, is summarised in Algorithm 1.

Algorithm 1: Construction of the PC-based chemical stressor matrix

Data: Site-by-chemical matrix $\mathbf{M}_{310\times16}$.

Result: PC-based chemical stressor matrix $\mathbf{S}_{310\times p}$.

1 Log-transform and standardise, then form the correlation matrix:
$\mathbf{M}_{310\times16} \to \mathbf{M}'_{310\times16} \xrightarrow{\text{correlation}} \mathbf{cor}_{16\times16}$

2 Apply PCA, quartimax rotation (flip if needed), and loading selection:
$\mathbf{cor}_{16\times16} \to \mathbf{Loadings}_{16\times p'} \xrightarrow{\text{rotation}} \mathbf{Loadings}^{\star}_{16\times p'} \xrightarrow{\text{selection}} \mathbf{Loadings}^{\star}_{16\times p}$

3 Project the processed chemical matrix onto the selected rotated loading space: $\mathbf{S}_{310\times p} = \mathbf{M}'_{310\times16}\,\mathbf{Loadings}^{\star}_{16\times p}$

4 **return** $\mathbf{S}_{310\times p}$

1.2.2 Latent Stressors and Contamination Cumulative Impact

In multiple-stressor ecology, stressors are commonly defined externally as ecologically interpretable pressures, such as nutrient enrichment, habitat degradation, flow alteration, or pollution. Studies such as those of Halpern et al. and Birk et al. adopt this perspective to examine how multiple recognisable environmental pressures combine, dominate, or interact in shaping biological responses (Halpern et al., 2008 [8]; Birk et al., 2020 [3]).

In the present study, by contrast, the chemical stressors are not directly defined pressure categories. They are latent contamination gradients extracted through the procedure of the preceding subsection, and are best interpreted as contamination syndromes (Huang et al. [9] and Loska and Wiechuła [11]). The latent stressors here therefore do not denote individual pollutants, but rather structured chemical pressures potentially experienced by the benthic communities.

From these latent stressors, an integrated contamination stress is derived for each site. We adopt the perspective that the community-level effects of the latent stressors are *additive*: their individual stress contributions are summed to yield a single integrated contamination stress. This quantity is denoted the Sum Relative Stress (SumRel), which expresses the position of a site relative to all other sites within the summed-stressor value space. Accordingly, the SumRel of site i is the sum of its stressor values across the p latent stressors. These values are the raw principal-component scores of the log-transformed, standardised chemical data $\mathbf{M}'$ projected onto the retained loadings, sign-aligned to a common direction in which higher scores indicate higher chemical concentrations.

$$\mathrm{SumRel}_i = \sum_{j=1}^{p} s_{ij}, \qquad s_{ij} \in \mathbf{S}_{310\times p},$$

where s_{ij} is the direction-aligned raw stressor value of site i on latent stressor j. Different contamination–community response perspectives may legitimately coexist. The additive view adopted here assumes that summing orthogonal stress values reflects the combined influence of the stressors on community composition (Halpern et al., 2008 [8]), whereas a dominance view holds instead that a single prevailing stressor obscures the others and governs the response, so that the integrated effect is set by the highest stressor value alone (Birk et al., 2020 [3]). We construct the integrated stress measure (SumRel) under the additive perspective, while acknowledging that the choice among such perspectives is one that ultimately be left to ecologists to judge.

1.2.3 Least-Contaminated Sites Selection and Validation

Proceeding from the integrated stress level, we define the *least-contaminated sites* as those lying at the low tail of the observed distribution of integrated contamination stress (SumRel). These sites constitute the least-contaminated group, whose taxa communities are expected to be the least affected by contamination and thus to serve as a benchmark, or baseline.

Determining this group generally requires a threshold to separate the least-contaminated sites from the remainder. In the study of Zhang (Zhang, 2008 [17]), this threshold was set at the 20th percentile of integrated stress. Such a 20% cut-off is, in most cases, chosen empirically (Australian and New Zealand Governments, 2018 [2]; Ministry for the Environment [12]). While this approach does capture the sites at the low end of the observed stress distribution, it neither examines how a fixed-percentile choice

(20%) weakens the least-contamination assumptions underlying the delineated group size when applied across differing contamination settings (Stoddard et al., 2006 [13]), nor verifies that a low-percentile group of a given size exerts a significantly weaker influence on community composition than other groups of the same size.

Prescribing an absolute count threshold (for instance, 50 or 60 sites) applicable across all contamination settings is likewise neither attainable nor interpretable, since such a number depends on the distribution of stress levels in the study context and on the researcher's judgement of what constitutes, and how to constitute, an ecologically "low" level of contamination.

1.2.4 Between-group resampling test

In this section we continue to adopt the 20th percentile to select the least-contaminated sites, but design a hypothesis test, based on a multivariate-regression resampling procedure, to verify whether the level of contamination impact on the community within the least-contaminated group is significantly lower than that of other, randomly drawn groups. The method draws on the first half of Redundancy Analysis (RDA), measuring the magnitude of contamination impact on the community within each group by fitting a multivariate regression model.

Specifically, for a given set of m sites with retained latent-stressor matrix $\mathbf{X}$ and community composition matrix $\mathbf{Y}$, we regress each species record $\mathbf{y}$ in $\mathbf{Y}$ on $\mathbf{X}$. The fitted records of all species are collected into $\hat{\mathbf{Y}}$, a matrix of the same dimensions as $\mathbf{Y}$. The variation in $\hat{\mathbf{Y}}$ is constrained by $\mathbf{X}$ and is taken as a measure of the community variation induced by the latent stressors that $\mathbf{X}$ represents. The adjusted coefficient of determination $\bar{R}^2$ of the model, that is, the adjusted ratio $\frac{\text{variation of } \hat{\mathbf{Y}}}{\text{variation of } \mathbf{Y}}$, expresses the proportion of total community variation that is accounted for under the constraint of the stressors, in other words the magnitude of contamination impact on the community. The exact computation of $\bar{R}^2$ from the regression fit is given in Appendix 1.5.2.

In this test, we randomly draw B site sets, each of size m_{ref}, denoted $\mathbf{S}_b$, $b \in \{1, \ldots, B\}$. The above multivariate regression model is fitted on each set, yielding a distribution of B values $\bar{R}_b^2$ that reflect the magnitude of contamination impact on the community under random sampling. The set of least-contaminated sites comprising the lowest 20th percentile is denoted $\mathbf{S}_{\mathrm{ref}}$; fitting the same model on this set yields a single value $\bar{R}^2_{\mathrm{ref}}$, reflecting the magnitude of contamination impact on the community within the least-contaminated set.

We then compare $\bar{R}^2_{\mathrm{ref}}$ against the B values $\bar{R}^2_b$ obtained by random resampling. If $\bar{R}^2_{\mathrm{ref}}$ is significantly lower than the randomly distributed $\bar{R}^2_b$ values, the level of contamination impact on the community within the least-contaminated set is taken to be significantly lower than that observed under random sampling, thereby validating that contamination stress exerts a lower level of influence on community composition within the least-contaminated set.

The specific computational procedure is summarised in Algorithm 2.

Algorithm 2: Between-group resampling test for reduced contamination impact in least-contaminated sites

Data: Latent-stressor matrix $\mathbf{X}$, community matrix $\mathbf{Y}$, reference size m_{ref}, replicates B.

Result: Significance of reduced contamination impact in $\mathbf{S}_{\mathrm{ref}}$.

Fit the reference set: take $\mathbf{S}_{\mathrm{ref}}$ as the lowest 20% of sites by SumRel, regress $\mathbf{Y}$ on $\mathbf{X}$, and record $\bar{R}^2_{\mathrm{ref}}$

for $b \leftarrow 1$ **to** B **do**

  Draw a random site set $\mathbf{S}_b$ of size m_{ref}, fit the same regression, and record $\bar{R}^2_b$

Compare $\bar{R}^2_{\mathrm{ref}}$ with the null distribution $\{\bar{R}^2_b\}_{b=1}^{B}$

return $\mathbf{S}_{ref}$ *is validated if* $\bar{R}^2_{ref}$ *lies significantly below* $\{\bar{R}^2_b\}$

This comparison constitutes a nonparametric, resampling-based hypothesis test. Rather than referring $\bar{R}^2_{\mathrm{ref}}$ to a theoretical sampling distribution, we construct an empirical null distribution $\{\bar{R}^2_b\}_{b=1}^{B}$ directly by random resampling, so that no parametric assumptions on the distribution of the test statistic are required. The hypotheses are stated in Table 1.1. Under H_0, the least-contaminated set behaves no differently from an arbitrary set of the same size, and $\bar{R}^2_{\mathrm{ref}}$ is an ordinary draw from the null distribution; under the one-sided alternative H_A, the least-contaminated set exhibits weaker stressor-constrained variation, so that $\bar{R}^2_{\mathrm{ref}}$ falls in the lower tail of $\{\bar{R}^2_b\}$. The corresponding left-tailed empirical p-value is

$$p \;=\; \frac{1 + \#\{\, b : \bar{R}_b^2 \le \bar{R}_{\mathrm{ref}}^2 \,\}}{B+1},$$

and H_0 is rejected at level α when $p \le \alpha$, validating that contamination stress exerts a significantly weaker influence on community composition within the least-contaminated set than under random sampling.

Table 1.1: Hypotheses for the resampling-based test of reduced stressor-constrained variation in the least-contaminated set.

H_i	**Statement**
H_0	The m_{ref} lowest-scored sites do not show weaker stressor-constrained variation than a random m_{ref}-site subset.
H_A	The m_{ref} lowest-scored sites show weaker stressor-constrained variation than a random m_{ref}-site subset.

1.2.5 Within-group permutation test

The between-group test above asks whether the least-contaminated set stands out *relative to other sets of equal size*. A complementary question can be posed about a single set in isolation: judged only against its own data, does the set carry a real stressor–community association, or none at all? We address this with a within-group permutation test.

For an observed set $\mathbf{S}_b$, we randomly permute the rows of its community matrix $\mathbf{Y}_b$ a total of B' times. Each permutation breaks the site-wise correspondence between stressors and community records, so the refitted value $\bar{R}_b^{2,\mathrm{perm}}$ measures the stressor-constrained variation attainable by chance alone. The B' permutations form a null distribution against which the observed $\bar{R}_b^2$ is judged.

For the least-contaminated set, we are interested in the opposite outcome to a conventional significance test: the set qualifies as a natural benchmark precisely when its stressor–community association cannot be distinguished from chance. If the observed $\bar{R}_b^2$ does *not* lie in the upper tail of the permutation distribution—that is, if it is no more extreme than the constrained variation produced by random row reassignment—then the contamination gradient exerts no detectable constraint on community composition within the set. The corresponding right-tailed permutation p-value,

$$p_b \;=\; \frac{1 + \#\{\, j : \bar{R}^{2,\mathrm{perm}}_{b,j} \geq \bar{R}^2_b \,\}}{B' + 1},$$

is the probability of obtaining an association at least as strong as the observed one when stressors and community are in truth unrelated. A *large* p_b therefore indicates that the observed association is indistinguishable from random reassignment, so that we can **not** reject the null hypothesis: the set carries essentially no contamination-constrained structure. Ecologically, failing to reject the null hypothesis is desirable: it characterises a community organised independently of the contamination gradient —whereas a small p_b would instead reveal a set within which contamination shapes community composition. The hypotheses tested here can be expressed in the form of Table 1.2, and the detailed computation can be summarised in the form of Algorithm 3.

Table 1.2: Hypotheses for the within-group permutation test of stressor-constrained variation. A least-contaminated set is expected to *fail to reject* H_0, indicating no detectable contamination constraint on community composition.

H_i	**Statement**
H_0	Within $\mathbf{S}_b$, the stressors constrain community variation no more strongly than under random row reassignment of $\mathbf{Y}_b$.
H_A	Within $\mathbf{S}_b$, the stressors constrain community variation significantly more strongly than under random row reassignment of $\mathbf{Y}_b$.

Algorithm 3: Within-group permutation test of stressor–community association

Data: Site set $\mathbf{S}_b = (\mathbf{X}, \mathbf{Y}_b)$, permutations B'.

Result: Significance of the stressor–community association within $\mathbf{S}_b$.

Fit the observed set: regress $\mathbf{Y}_b$ on $\mathbf{X}$ and record $\bar{R}_b^2$
for $j \leftarrow 1$ **to** B' **do**
    Permute the rows of $\mathbf{Y}_b$ to obtain $\mathbf{Y}_b^{\text{perm}}$, refit on $\mathbf{X}$, and record $\bar{R}_{b,j}^{2,\text{perm}}$
Compute the right-tailed p-value p_b from $\bar{R}_b^2$ and $\{\bar{R}_{b,j}^{2,\text{perm}}\}_{j=1}^{B'}$
return $\mathbf{S}_b$ *behaves as a natural benchmark if* $p_b > \alpha$ *(the contamination constraint is indistinguishable from random reassignment)*

Together, the between-group resampling test of Algorithm 2 and the within-group permutation test of Algorithm 3 probe the least-contaminated set from two complementary angles—its standing relative to comparable sets, and the strength of its own internal stressor–community signal. By scanning candidate sets of increasing size ordered by SumRel and tracking the size at which p_b falls below α, we can observe how the size of least-contaminated group affects the testing outcomes and have insights for choosing a defensible group size.

The scanning procedure is an extension of Algorithm 2 and 3, increasing the least-contaminated group size from 15 to 310 sites in stepwise increments of 5 sites and performing the two tests at each step. At every size m_{ref}, the reference curves $\bar{R}_{\text{ref}}^2$ and p_{ref} are recorded together with the random-subset bands of the same statistics, as summarised in Algorithm 4.

After we apply this procedure across cases, it will show an informative result: in the *corridor case*, where lightly contaminated sites are comparatively abundant, it supports a relatively large least-contaminated set, so that the 20% threshold is a reasonable choice; in the *Detroit River case*, where such sites are scarcer, it favours a smaller set, so that the same 20% threshold becomes difficult to justify.

Algorithm 4: Threshold scan across least-contaminated subset sizes

Data: Latent-stressor matrix $\mathbf{X}$, community matrix $\mathbf{Y}$, SumRel-ordered sites, step $\delta = 5$.

Result: Curves $\bar{R}^2_{\mathrm{ref}}(m_{\mathrm{ref}})$ and $p_{\mathrm{ref}}(m_{\mathrm{ref}})$ with random-subset bands.

for $k \leftarrow 0$ **to** 59 **do**
- $m_{\mathrm{ref}} \leftarrow 15 + k\,\delta$
- Take $\mathbf{S}_{\mathrm{ref}}$ as the m_{ref} lowest-SumRel sites, and record $\bar{R}^2_{\mathrm{ref}}$ and p_{ref} with their random-subset bands (Algorithms 2–3)

return $\{\bar{R}^2_{ref}(m_{ref})\}$ *and* $\{p_{ref}(m_{ref})\}$

1.3 Results

1.4 Tables and Figures

Table 1.3: Rotated PCA loadings and variance summary for the first six principal components

Chemicals	**PC1**	**PC2**	**PC3**	**PC4**	**PC5**	**PC6**
Cr	**0.9174**	0.2387	0.1500	-0.0043	-0.0663	0.0623
Zn	**0.8865**	-0.0024	-0.0226	-0.0688	0.1449	0.0614
Ni	**0.8861**	0.0193	-0.0291	0.0311	-0.2636	-0.0020
Cu	**0.8839**	0.2188	0.1136	-0.0860	-0.0045	-0.0379
total PCB	**0.8364**	-0.3031	0.0049	0.1665	0.1523	0.0210
Cd	**0.7387**	-0.0877	0.4223	0.0595	0.1766	0.1071
Co	**0.6772**	0.6583	0.0279	0.0688	0.1882	0.0637
p,p'-DDE	**0.6385**	-0.1622	-0.1223	0.4012	0.2385	0.0789
Mn	**0.5012**	0.3860	0.3665	0.1539	-0.3701	0.1585
Ca	**0.2516**	0.1462	0.0287	0.1107	0.0568	0.0954
Al	0.4609	**0.8085**	0.0437	0.0214	0.1118	0.1404
Pb	0.3790	-0.0207	**0.8037**	0.0162	-0.3245	-0.0084
As	0.3145	0.1097	**0.7899**	-0.0134	0.3535	0.0510
OCS	0.1816	0.0541	0.0238	**0.9356**	0.0096	0.0095
Hg	0.4581	0.2655	0.0210	0.0609	**0.7510**	0.0320
Fe	0.4722	0.1712	0.0579	0.0212	0.0190	**0.8420**
Explained Var.	6.5407	1.6036	1.6422	1.1264	1.1890	0.7980
Prop. Var.	0.4088	0.1002	0.1026	0.0704	0.0743	0.0499
Cum. Prop.	0.4088	0.5090	0.6117	0.6821	0.7564	**0.8063**

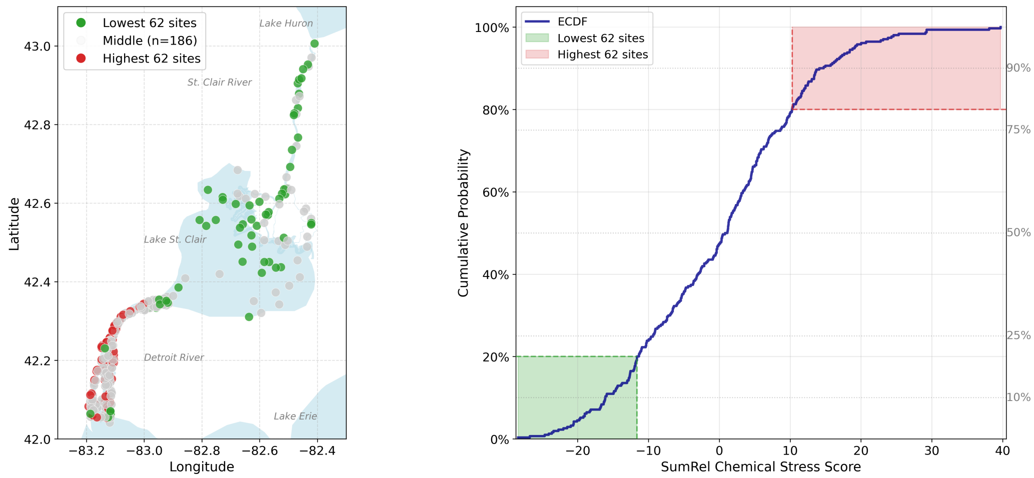

Figure 1.1: Sum Relative Contamination Score (SumRel) across the 310 corridor sites. Left: corridor map of the lowest 20% (62 sites, green), the highest 62 sites (red), and the middle 186 sites (grey). Right: SumRel ECDF, with the lowest and highest 62 sites bifurcated at the 20th and 80th percentiles.

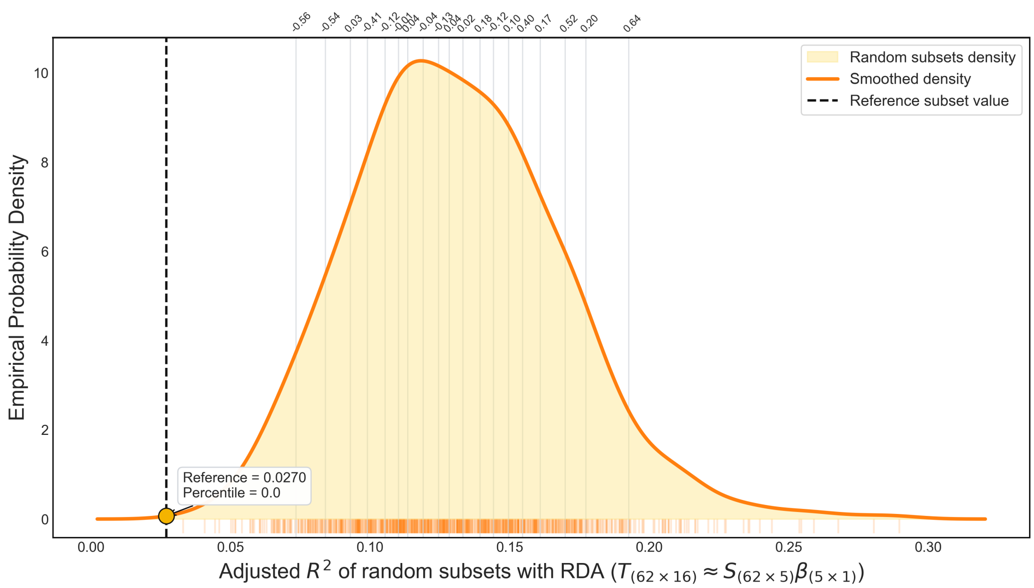

Figure 1.2: Between-group resampling test (Algorithm 2) at $m_{\text{ref}} = 62$: empirical $\bar{R}^2$ distribution of 1000 random 62-site subsets. The top axis shows average contamination scores of the 50 nearest subsets per location, with tick density proportional to probability mass. The black line marks the reference $\bar{R}^2$ (percentile = 0.0); with $p \leq 0.05$, H_0 is rejected—the least-contaminated set is significantly cleaner than random sets.

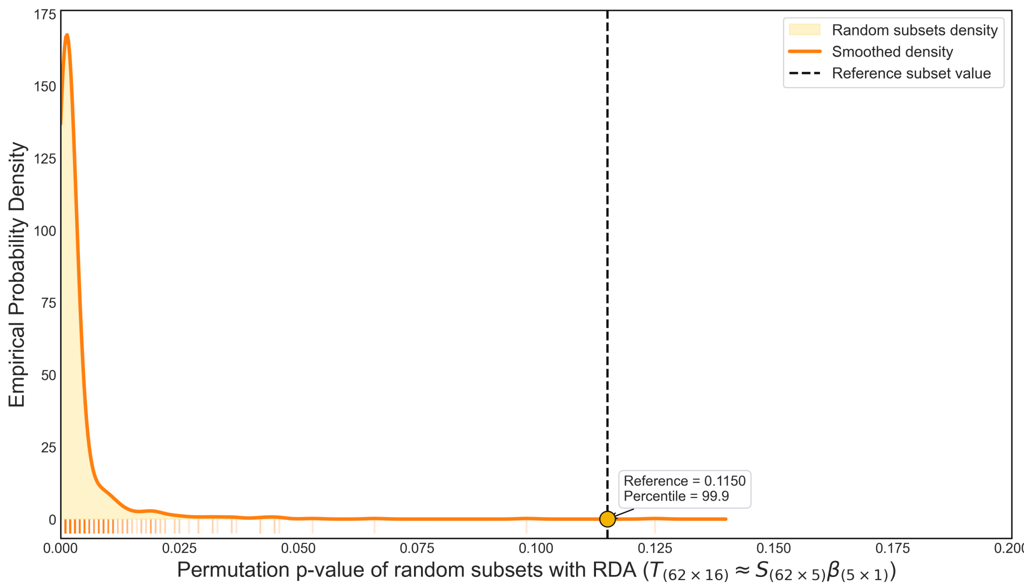

Figure 1.3: Within-group permutation test (Algorithm 3) at $m_{\text{ref}} = 62$: empirical distribution of permutation p-values from 1000 resampled subsets, each permuted 1000 times. The black line marks the reference p-value (percentile = 99.9); with $p \geq 0.05$, H_0 is not rejected—within the least-contaminated set the constraint is indistinguishable from chance, as expected of a natural benchmark.

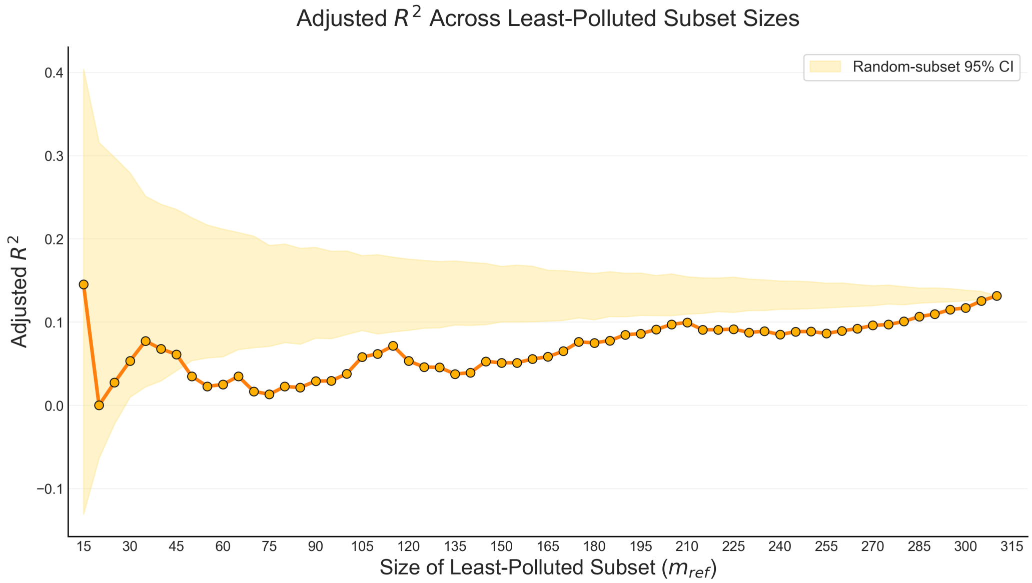

Figure 1.4: Threshold scan of $\bar{R}^2$ across subset sizes (Algorithm 4). The orange curve traces $\bar{R}^2_{\mathrm{ref}}$ of the m_{ref} lowest-SumRel sites for $m_{\mathrm{ref}} = 15$ to 310; the band is the random-subset 95% interval. The curve hugs the lower edge at small sizes and rises toward the band centre as more contaminated sites are admitted.

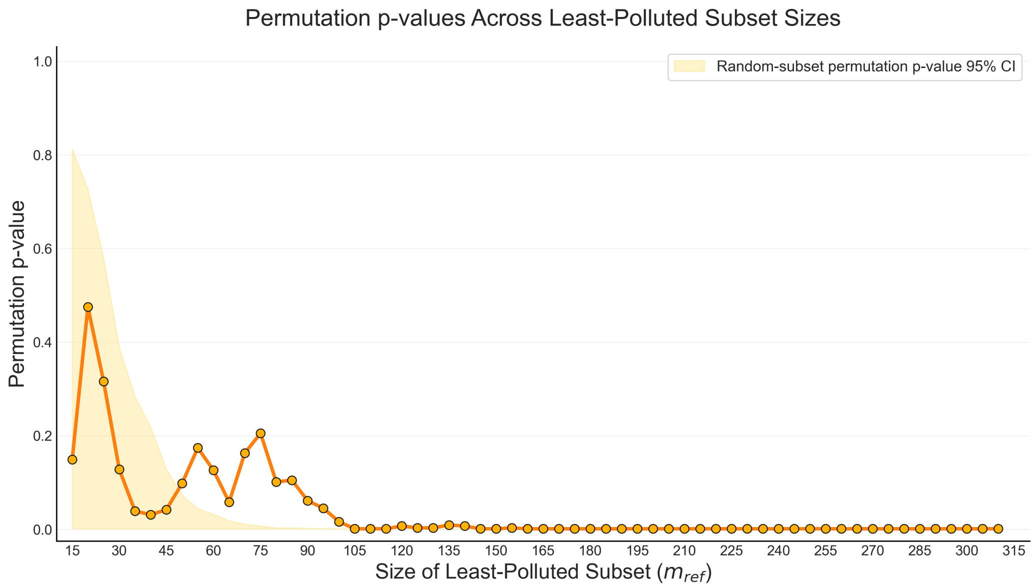

Figure 1.5: Threshold scan of the permutation p-value across subset sizes (Algorithm 4). The orange curve traces p_{ref} of the m_{ref} lowest-SumRel sites for $m_{\text{ref}} = 15$ to 310; the band is the random-subset 95% interval. The reference p_{ref} stays high for small sets and collapses toward zero beyond $m_{\text{ref}} \approx 100$, marking where contamination begins to constrain community composition.

1.5 Appendix

1.5.1 Orthogonality and variance preservation of principal components

Setup. Let $\mathbf{X}$ be an $n \times p$ data matrix recording p descriptors (variables) over n objects (sites). Assume each column has been centred and standardised to unit variance, so that the $p \times p$ correlation matrix of the column space is

$$\mathbf{R} \;=\; \tfrac{1}{n-1}\,\mathbf{X}^{\top}\mathbf{X}.$$

By construction $\mathbf{R}$ is *symmetric* ($\mathbf{R}^{\top} = \mathbf{R}$) and *positive semi-definite*, since for any vector $\mathbf{u} \in \mathbb{R}^p$,

$$\mathbf{u}^{\top}\mathbf{R}\,\mathbf{u} = \tfrac{1}{n-1}\,(\mathbf{X}\mathbf{u})^{\top}(\mathbf{X}\mathbf{u}) = \tfrac{1}{n-1}\,\|\mathbf{X}\mathbf{u}\|^2 \;\geq\; 0.$$

Step 1 — Eigen-decomposition exists and yields an orthonormal basis. By the spectral theorem, every real symmetric matrix admits a decomposition

$$\mathbf{R}\,\mathbf{v}_k = \lambda_k\,\mathbf{v}_k, \qquad k = 1, \dots, p,$$

with *real* eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$ (non-negative by positive semi-definiteness). The eigenvectors are mutually orthogonal: for $\lambda_i \neq \lambda_j$,

$$\lambda_i\,\mathbf{v}_i^\top \mathbf{v}_j = (\mathbf{R}\mathbf{v}_i)^\top \mathbf{v}_j = \mathbf{v}_i^\top \mathbf{R}\mathbf{v}_j = \lambda_j\,\mathbf{v}_i^\top \mathbf{v}_j \implies (\lambda_i - \lambda_j)\,\mathbf{v}_i^\top \mathbf{v}_j = 0 \implies \mathbf{v}_i^\top \mathbf{v}_j = 0.$$

Hence we may normalise them into an orthonormal set, collected as columns of an orthogonal matrix $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_p]$ with $\mathbf{V}^\top\mathbf{V} = \mathbf{V}\mathbf{V}^\top = \mathbf{I}$, giving

$$\mathbf{R} = \mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{V}^\top, \qquad \boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \dots, \lambda_p).$$

Step 2 — The eigenvectors preserve the structure of R. Because $\mathbf{V}$ is orthogonal, the decomposition is an *exact* change of basis: no information in $\mathbf{R}$ is lost, only re-expressed. The eigenvectors form the directions along which $\mathbf{R}$ acts purely by scaling, and the total variance is conserved,

$$\mathrm{tr}(\mathbf{R}) = \mathrm{tr}(\mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^\top) = \mathrm{tr}(\boldsymbol{\Lambda}) = \sum_{k=1}^{p} \lambda_k = p,$$

so each λ_k is precisely the amount of total variance carried by direction $\mathbf{v}_k$. The structure of $\mathbf{R}$ is thus repackaged, without distortion, into ordered orthogonal axes.

Step 3 — Projection gives orthogonal scores that retain maximal variation. Project the descriptors onto the eigenvectors to obtain the principal-component scores

$$\mathbf{F} = \mathbf{X}\,\mathbf{V}, \qquad \mathbf{f}_k = \mathbf{X}\,\mathbf{v}_k.$$

These new axes are *mutually uncorrelated*: their covariance matrix is diagonal,

$$\tfrac{1}{n-1}\,\mathbf{F}^\top\mathbf{F} = \mathbf{V}^\top\Big(\tfrac{1}{n-1}\mathbf{X}^\top\mathbf{X}\Big)\mathbf{V} = \mathbf{V}^\top\mathbf{R}\,\mathbf{V} = \boldsymbol{\Lambda},$$

so $\mathrm{Var}(\mathbf{f}_k) = \lambda_k$ and $\mathrm{Cov}(\mathbf{f}_i, \mathbf{f}_j) = 0$ for $i \neq j$. Because the λ_k are ordered, the first few components carry the largest share of variation: retaining the leading $m < p$ axes captures the fraction

$$\frac{\sum_{k=1}^{m} \lambda_k}{\sum_{k=1}^{p} \lambda_k}$$

of the total variance, the maximum attainable by any m orthogonal directions (Rayleigh–Ritz). The optionally *scaled* eigenvectors $\mathbf{v}_k\sqrt{\lambda_k}$ (distance/correlation biplot scaling) rescale axis length to reflect λ_k while leaving orthogonality untouched.

1.5.2 Computation of the adjusted constrained variation $\bar{R}^2$

Setup. Consider a site set of m objects with latent-stressor matrix $\mathbf{X} \in \mathbb{R}^{m\times p}$ (an intercept column prepended) and Hellinger-transformed, column-centred community matrix $\mathbf{Y} \in \mathbb{R}^{m\times q}$ over q taxa. The aim is to measure how much community variation the stressors constrain.

Step 1 — Multivariate least-squares fit. Regressing all taxon columns of $\mathbf{Y}$ on $\mathbf{X}$ gives the fitted values

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}, \qquad \mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top,$$

where $\mathbf{H}$ projects onto the column space of $\mathbf{X}$; $\hat{\mathbf{Y}}$ is thus the portion of the community response lying within the stressor space.

Step 2 — Variation partitioning. With $\mathbf{H}$ symmetric and idempotent, the explained fraction of community variation is

$$R^2 = \frac{\mathrm{SS}_{\mathrm{fit}}}{\mathrm{SS}_{\mathrm{tot}}} = \frac{\mathrm{tr}(\mathbf{Y}^\top\mathbf{H}\mathbf{Y})}{\mathrm{tr}(\mathbf{Y}^\top\mathbf{Y})}.$$

Step 3 — Adjustment for dimensionality. Since R^2 rises with p and falls with m, sets of differing size are not directly comparable; Ezekiel's degrees-of-freedom correction (Ezekiel, 1941 [6]) gives

$$\bar{R}^2 = 1 - \left(1 - R^2\right)\frac{m-1}{m-p-1}.$$

This $\bar{R}^2$ is the statistic compared across reference, random, and permuted sets of common size m_{ref}, so that differences reflect the strength of the stressor–community constraint rather than the dimensions of the fit.

Chapter 2

Natural Community Types: Partitioning Natural Variation and Associating with Environmental Descriptors

2.1 Introduction

Before the contamination stress influence on the community can be examined in isolation, the influence of the natural environment must be taken into account. The community records of interest at any point in space and time are the joint outcome of natural environmental conditions, contamination pressure, and other, unrecorded sources of stress(Allan, 2004 [1]). Therefore a site's taxa record is a representation of all these factors acting together.

Two strategies are commonly used: adjusting for the natural environment within the model, or stratifying the data by it before the model is fitted. The first introduces the environmental variables as covariates in the regression of the community records of interest, so that contamination and natural conditions are modelled jointly and estimated finely; its drawback is that contamination gradients typically covary with natural environmental gradients (Graham, 2003 [7]), resulting collinearity with unstable coefficients and limited practical value. The second strategy instead treats the natural environment as a categorical descriptor and partitions the sites into classes of similar environmental and(or) regional—characteristics (Hughes, Larsen, and Omernik, 1986 [10]; Allan, 2004 [1]) to avoid introducing covariates. This stratification can be a more interpretable approach, since it isolates the contamination signal within definable and comparable habitat stratas and divides the community records of interest with natural variation into community types with aligned environmental conditions.

However, the difficulty is practical rather than conceptual. Natural community composition is shaped by a large set of environmental features, many of which are unobservable, observable but not recorded, or recorded only with substantial measurement uncertainty. Consequently, in most empirical settings, the limited set of measured environmental variables is insufficient to define site groups that are genuinely homogeneous in both environmental conditions and the community records of interest. This limitation is illustrated in Figure 2.1. Complete environmental observation aligns the partition in Ω_{env} with the true community boundary; incomplete observation yields only a degraded partition that mislabels the sites between it and the true boundary—an error of incomplete descriptors, not of a failed environment–community relationship.

For this reason, we reverse the order in which environment and community enter the stratification. We first let the community data *define* the types directly, and then ask what environmental signature *describes* them. It rests on the previous conclusion: within the least-contaminated set, the recorded assemblages retain natural community composition. Therefore we can read natural community types from these least-contaminated sites, since they carry natural variation that we wish to stratify by. Starting with the least-contaminated set, we cluster its community matrix, validate and retain K branches as natural community types. Next, we model this environment-assemblage association by training a classifier that maps their environmental descriptors onto the defined community types. Through this process, we uncanonically stratify community types on the taxa space Ω_{taxa}, makes the implicit environment-determination assumption become an explicit and measurable property in the classifier building. This *define*-then-*describe* workflow is illustrated in Fig-

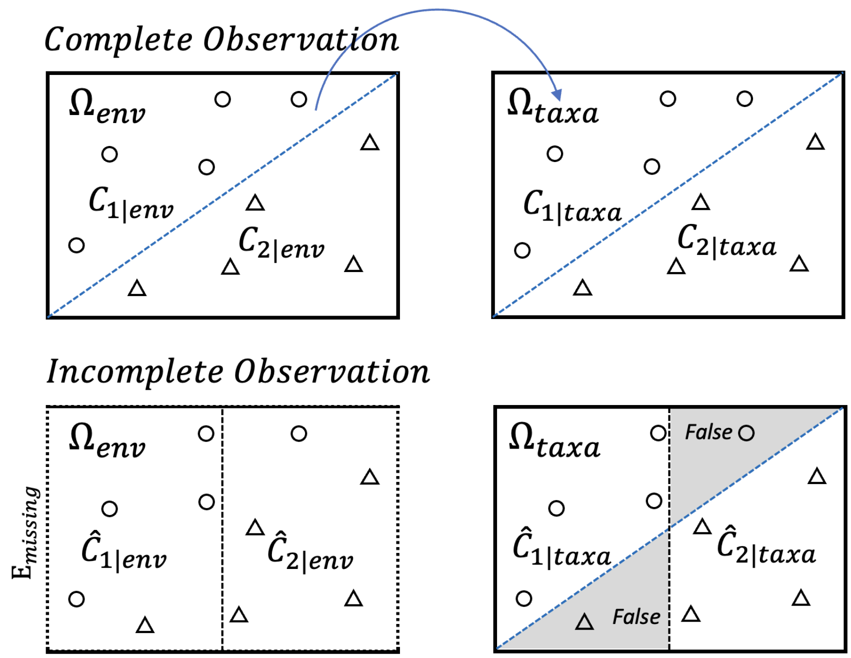

Figure 2.1: Under the assumption that environment determines community, incomplete environmental observation propagates classification error from Ω_{env} to Ω_{taxa}. Complete observation recovers the true diagonal boundary; incomplete observation imposes a degraded vertical partition, mislabelling sites in the wedge between the two (shaded, "False").

ure 2.2.

The built classifier is then applied to the remaining, contaminated sites: predicting their environmental descriptors into community types. Through this process, every site regardless of contamination status is assigned to its natural community type in a broad sense.

2.2 Methods

The overall aim of this stage is to assign every site, regardless of contamination level, to a natural community type. The procedure has three main steps, the first two carried out entirely within the least-contaminated set and the third applying their results to the remaining sites. First, natural community types are defined on the least-contaminated sites by clustering their community matrix. Second, starting from these

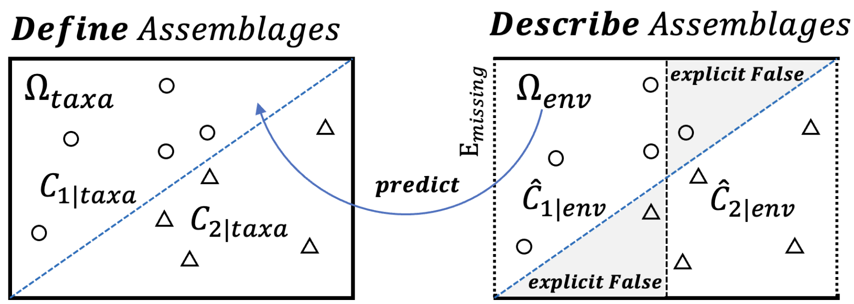

Figure 2.2: Community types are *defined* on the taxa space Ω_{taxa}, then *described* and predicted from the environmental space Ω_{env}. Limited environmental features make some misassignments, but these are rendered explicit (shaded, "explicit False").

defined types, a discriminating subset of environmental descriptors is identified and a classifier is built to quantify the relationship between environmental features and community types. Third, the classifier is applied to the remaining contaminated sites, predicting and delineating the decision boundary by which they are assigned to the corresponding natural community type. The remainder of this section describes each step in detail.

2.2.1 Unconstrained community typing and its statistical validation

Dissimilarity between sites, Double-zero problem and Hellinger transformation

Hierarchical clustering operates on a matrix of pairwise dissimilarities, so the first choice is how dissimilarity between sites is measured from the community records. Applying the Euclidean distance directly to raw abundances is inappropriate for community data, because it treats the joint absence of a taxon at two sites as evidence of similarity—the *double-zero problem*—and lets a few abundant taxa dominate the distance. To avoid this, the community matrix is first Hellinger-transformed: at a given site, each taxon's record is divided by the site's total and the square root is taken, so that the entry becomes the square root of the taxon's relative abundance at that site. The Euclidean distance computed between two Hellinger-transformed site profiles is then equal to the Hellinger distance between their original compositions, a measure that is insensitive to double zeros and to differences in total abundance, and which is recommended for ordination and clustering of community data in numerical ecology (Borcard, Gillet, and Legendre, 2018 [4]). This transform therefore makes the subsequent Euclidean-based clustering ecologically meaningful.

Hierarchical clustering with Ward's method

The Hellinger-based dissimilarities are clustered by agglomerative hierarchical clustering. Among the available linkage functions, Ward's method is chosen because it merges, at each step, the pair of clusters whose fusion produces the smallest increase in within-cluster variance of the Euclidean (here Hellinger) distances. This criterion suits the goal of delineating community types with members mutually similar in composition. The procedure yields a dendrogram in which the height of each fusion records the linkage function value across which the corresponding branches join; cutting the dendrogram at a chosen linkage height then defines the K community types retained for the subsequent analysis.

Statistical evidence for the branching: node-wise and overall ANOVA

A dendrogram will partition any data, so the branching structure requires statistical support. Two complementary tests are used, both applied to the Hellinger-transformed taxa abundances used in the clustering rather than to the raw counts. First, a *node-wise* ANOVA is applied from the first to $K + 2$ branching nodes of the dendrogram: for every taxon, its Hellinger-transformed abundances in the two daughter branches are compared, identifying the taxa whose mean values differ significantly across the split. These taxa are the indicator species that contribute most to the formation of that particular node and drive the bifurcation. Second, once the K community types are finalised, an overall ANOVA tests, for each taxon, whether its Hellinger-transformed mean abundance differs significantly among the K groups, providing statistical evidence that the retained partition separates the community types along interpretable compositional gradients.

Robustness of the partition with pvclust

Beyond the compositional evidence for each branch, the stability of the branches under resampling is assessed with pvclust (Suzuki and Shimodaira, 2006 [15]; Suzuki, Terada, and Shimodaira, 2019 [16]). The method repeatedly perturbs the descriptor space by multiscale bootstrap resampling of the taxa, re-runs the same clustering procedure on each resampled space, and checks whether each branch of the original (anchor) dendrogram reappears. From the frequency with which a branch recurs across the B bootstrap replicates, an *Approximately Unbiased* (AU) p-value is computed for that branch. A high AU p-value means the branch is recovered in a large proportion of the perturbed spaces—its existence does not depend on a few particular taxa but is supported by many. The ordinary *Bootstrap Probability* (BP), estimated from fixed-size bootstrap resampling alone, is also reported, but it is a more biased than the AU value. Branches with low AU support are treated as uncertain and inform the choice of where the dendrogram is cut. Figure 2.3 illustrates the computation workflow [1], more details about this procedure can be found in the articles by Suzuki and Shimodaira (Suzuki and Shimodaira, 2006 [15]).

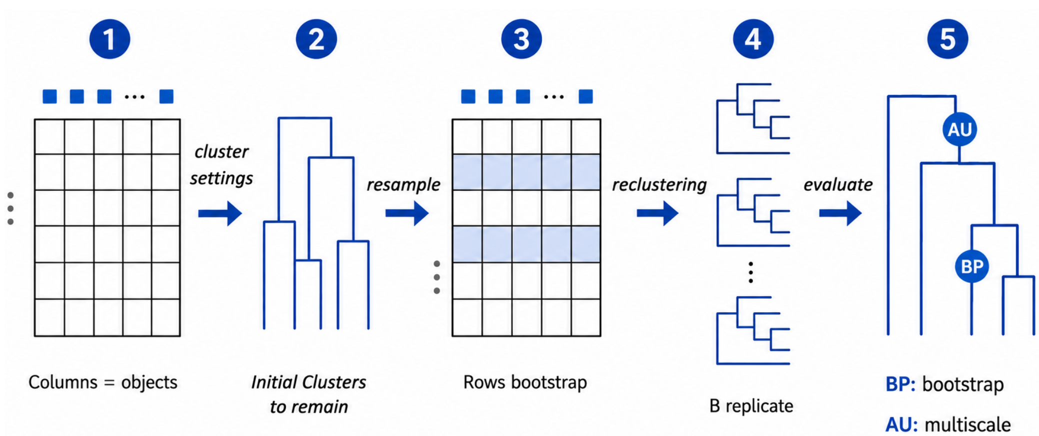

Figure 2.3: The pvclust workflow: bootstrap-resample taxa, recluster B times, and score each anchor branch by its AU (multiscale) and BP (bootstrap) recurrence.

[1] The data matrix is transposed so that taxa occupy the rows, because resampling routines in R and similar environments operate row-wise; bootstrapping the feature variables therefore requires placing them as rows.

2.2.2 Linking community types to environmental descriptors: classifier training under limited features

Limited environmental features and the 2008 reference-typing solution

A key difficulty of the present design persists even after the community types are defined: the limited set of measurable environmental features. Types delineated by unconstrained clustering may remain entangled in environmental space, where a missing key variable, or attention to uninformative ones, blurs the environment–type relationship. This already surfaced in the 2008 corridor study.

There, the Ward-clustered community types could not be cleanly separated on the five available environmental variables, so a directly trained classifier would have performed poorly. The solution was to define the two types by a selective assignment of branches. As shown in Figure 2.7, the dendrogram first divides into B_2 versus the pair $\{B_1, A\}$, and only at a lower height do B_1 and A separate. Cluster C_1 was assigned to B_2 and B_1 jointly, and Cluster C_2 to A alone—a definition under which the types separate more readily on the available variables.

The cost is that the types no longer reflect the dendrogram. The assignment is *non-monophyletic*: C_1 joins B_2 and B_1, two branches that do not form a single sub-tree, while the topologically closer A is placed in C_2. No single horizontal cut can produce this grouping; the labels are assigned across the tree's natural splitting order. The 2008 typing thus lets environmental separability override the community structure the clustering reveals, and the classifier's resulting accuracy is correspondingly inflated (Figure 2.8).

In the descriptor-space view, this operation amounts to relabelling the clustering-defined sites: circles (C_1) that sit, on the available features, among the triangles (C_2) are reassigned to C_2, buying environmental separability at the cost of the true taxa labels (Figure 2.4).

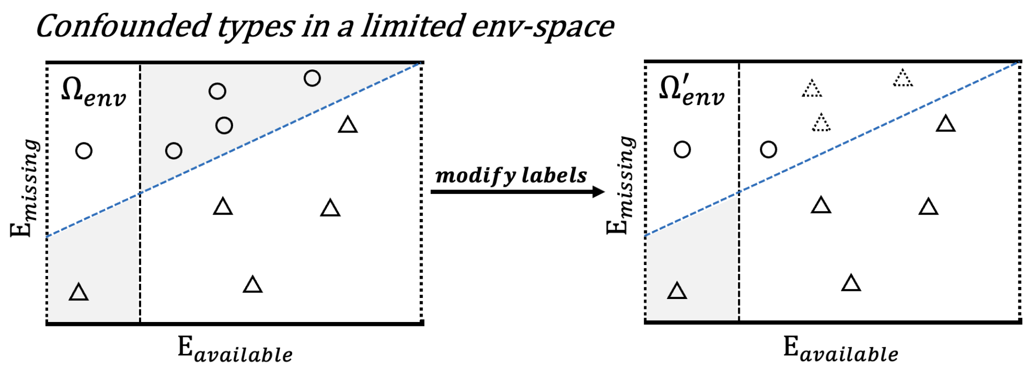

Figure 2.4: Confounded community types in a limited environmental space, illustrating the 2008 approach of Zhang. The true type boundary (blue dashed) is governed by an unobserved variable $E_{missing}$—its slope, rather than a diagonal, signifies that the missing variable is the more decisive one—so on the available features $E_{available}$ the types overlap (shaded). Zhang's solution relabels part of C_1 (circles) as C_2 (triangles) in one direction only ($\Omega_{env} \to \Omega'_{env}$, dotted triangles), reducing but not removing the overlap—separability bought by altering the true cluster labels.

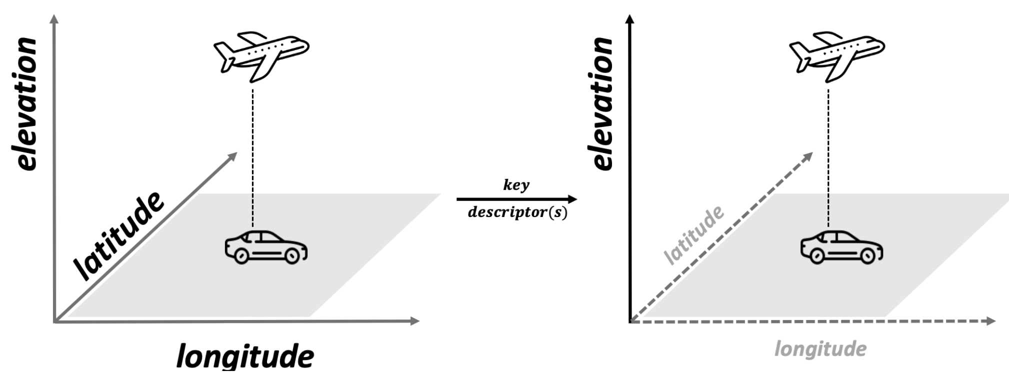

Figure 2.5: Not all descriptors help separate a small number of classes. An aircraft and a car are barely distinguishable on the horizontal descriptors (longitude, latitude), yet a single key descriptor—elevation—separates them cleanly. When only a few classes must be told apart, the right descriptor subset matters more than the number of descriptors, and retaining irrelevant ones only adds noise.

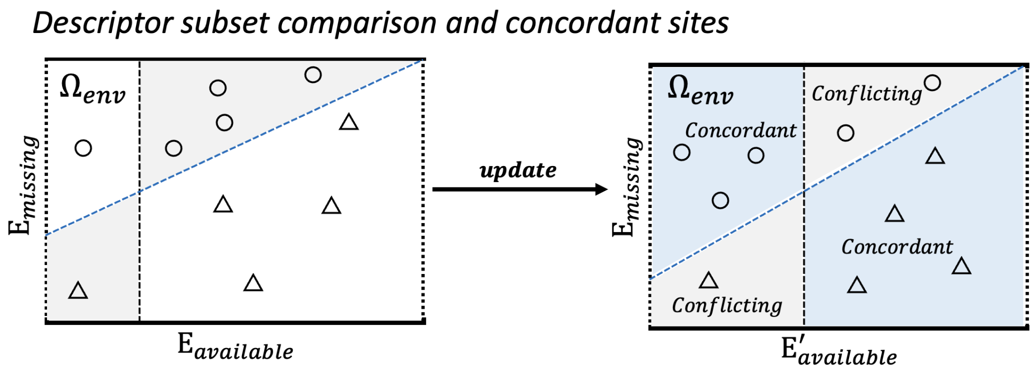

Figure 2.6: Selecting a descriptor subset and identifying concordant sites. On the initial descriptors $E_{available}$, the taxa-defined types overlap (left). Choosing a more discriminating subset $E'_{available}$ (right) separates most sites into *concordant* regions (blue), where taxa membership and environmental position agree, while the residual *conflicting* regions (grey) mark sites whose two views disagree. Only concordant sites are used to train the classifier; conflicting sites are retained as diagnostics. In practice, concordance is resolved across four categories (concordant, taxa-only, env-only, ambiguous), here simplified to two.

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2.2.3 Appendix

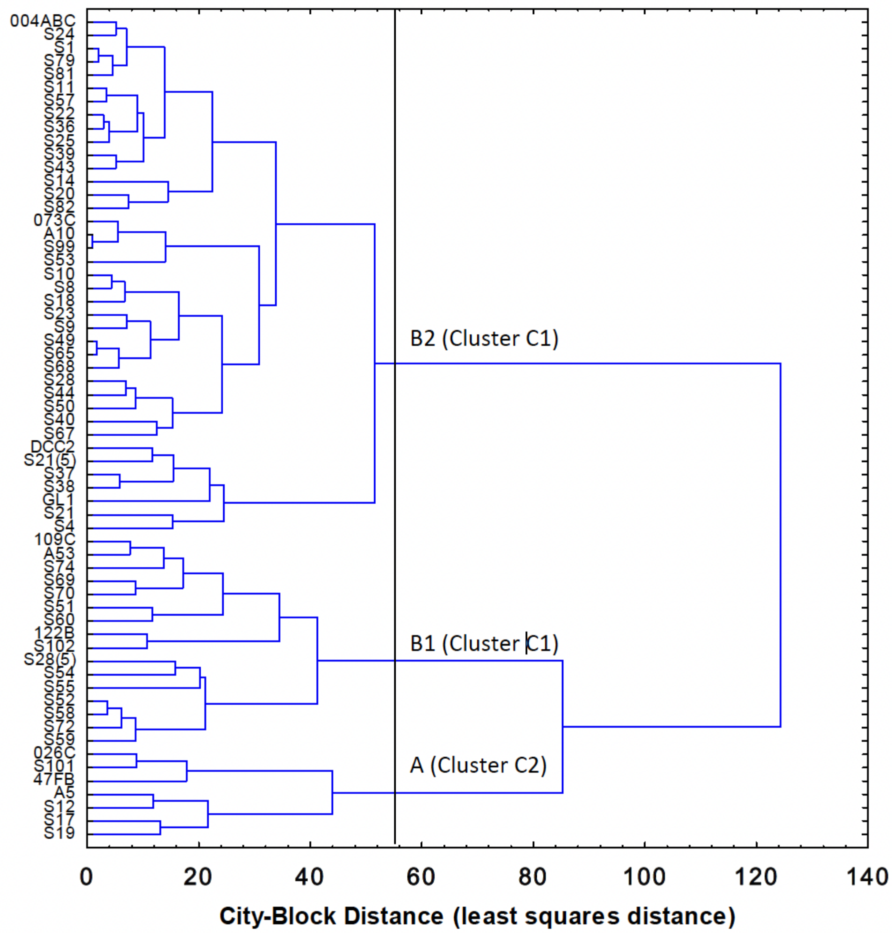

Figure 2 Dendrogram of REF sites (n = 62) grouped according to similar zoobenthic community composition in the 1991, 1999 and 2004/5 Lake Huron-Lake Erie Corridor analysis (Ward's method clustering city-block distances of octave-transformed relative abundances of zoobenthic taxa).

Figure 2.7: Branch-selective definition of community types in the 2008 reference dendrogram. The tree first separates B_2 from $\{B_1, A\}$, and only at a lower height splits B_1 from A. Cluster C_1 is assigned to B_2 and B_1 jointly, and Cluster C_2 to A alone (red boxes)—a non-monophyletic grouping that no single horizontal cut can produce, since A is topologically closer to B_1 than B_2 is.

Table 2.5. Summary of observed number of Lake Huron-Lake Erie Corridor sites in each cluster (columns) identified by zoobenthic taxa relative abundances and membership predicted (rows) by discriminant function classification (Appendix III) on the basis of habitat characteristics measured at those sites

		Observed	
Group	% Correct	Cluster C1	Cluster C2
Cluster C1	98	54	1
Cluster C2	71	2	5
Total	95	56	6

Figure 2.8: To do